

Protein sources

Chapter 3

Protein sources

- Proteins may be :
 1. Extracted from their naturally producing source material, such proteins are called '**homologous**' proteins
 2. Produced via recombinant DNA technology (recombinant protein production), often termed '**heterologous**' proteins.
- Advances in molecular biology and the 'omics' era rendered recombinant-based protein expression the more usual means of protein production.

Recombinant versus non-recombinant production

- Recombinant Protein production has a number of advantages over direct extraction from a native source material:
 1. Overcomes problems of source availability.
 2. Overcomes problems of source safety.
 3. Facilitates predefined modification of the protein's amino acid sequence.

Recombinant versus non-recombinant production

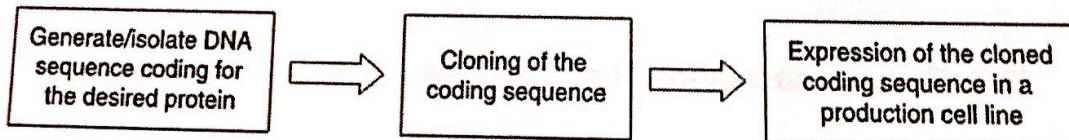
- Some proteins obtained by direct extraction from their native source for many reasons, including the following.
 1. Some proteins are produced by native production sources at very high levels.
 2. Some proteins, when expressed in recombinant systems, accumulate in a partially/fully unfolded, biologically inactive.
 3. Some proteins can be quite toxic to the recombinant cells in which they are being produced.
 4. Production in a recombinant system can influence specific post-translational modifications (PTMs) associated with a protein.
 5. Public resistance to the use of recombinant products for industrial applications (e.g. food processing),
 6. Patent issues surrounding the recombinant production route.

Approaches to recombinant protein production

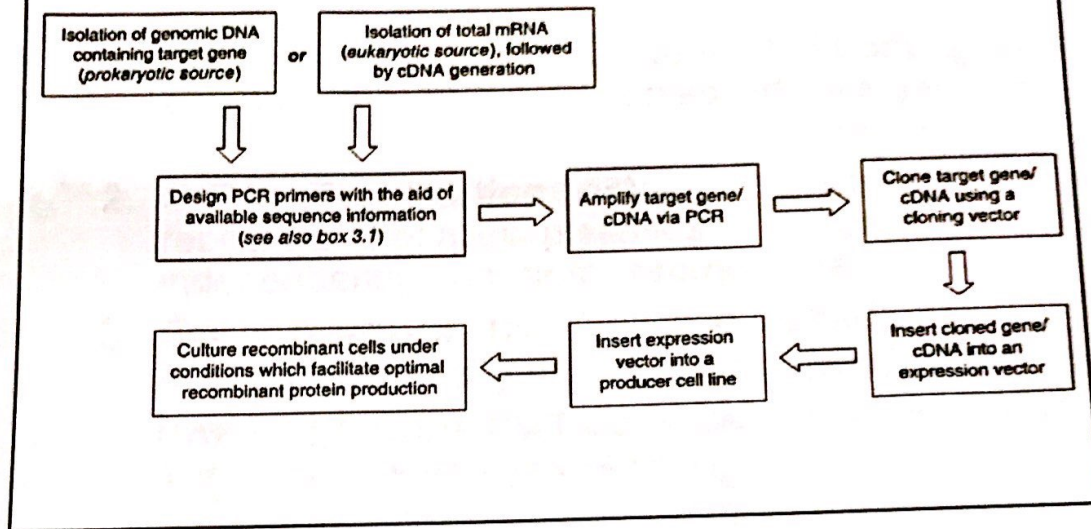
- A wide variety of molecular biological approaches may be pursued in order to construct a recombinant protein source organism. The specific approach chosen will depend on a number of considerations, including:
1. How much information is already known about the protein
 2. The ultimate use for which the protein is required (e.g. for research or for a specific commercial application)
 3. To some extent the personal preferences of the genetic engineers developing the recombinant system.

Approaches to recombinant protein production

(a)



Approaches to recombinant protein production



Cloning the target gene

- PCR allows the generation of multiple copies of the specific sequence encoding the target protein.
- The PCR product is then generally inserted into a **cloning vector**.
- The most commonly used vectors in conjunction with *E. coli* are plasmids.
- Plasmids are circular extrachromosomal DNA molecules and generally house one or more genes whose product renders the plasmid-containing cell resistant to specific antibiotic(s).
- One plasmid classically used in cloning experiments with *E. coli* is pUC18

Cloning vector features

• Practical Features of DNA Cloning Vectors:

1. **Size** – small enough to be separated from chromosomal DNA of host plasmid
 * inside bacteria: genomic DNA, plasmids
 * there are several methods to separate them, one of them is by size (the heavy goes down) by centrifugation
- * plasmids never integrate into the host chromosome
2. **Origin of replication (*ori*)** – site for DNA replication that allow plasmids to replicate independently from host chromosome
3. **Copy number:** number of plasmids in the cell (normally small but plasmids have high copy numbers) (more than one copy)
 (more copies = more production)

Cloning vector features

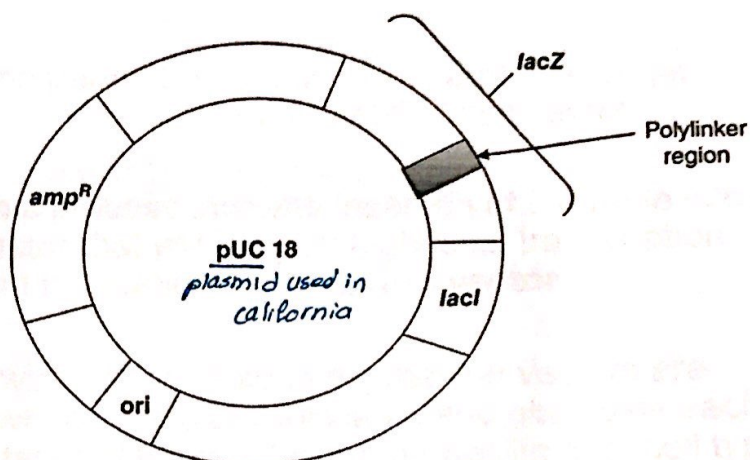
4. **Multiple cloning site (MCS)** – recognition sites for several restriction enzymes in which insert is cloned into (sticky ends)
 → because we might clone more than one gene

5. **Selectable marker genes** – allow to select for transformed colonies (inside the gene we'll have the cut)

6. **RNA polymerase promoter sequences** – used for transcription *in vitro* and *in vivo*

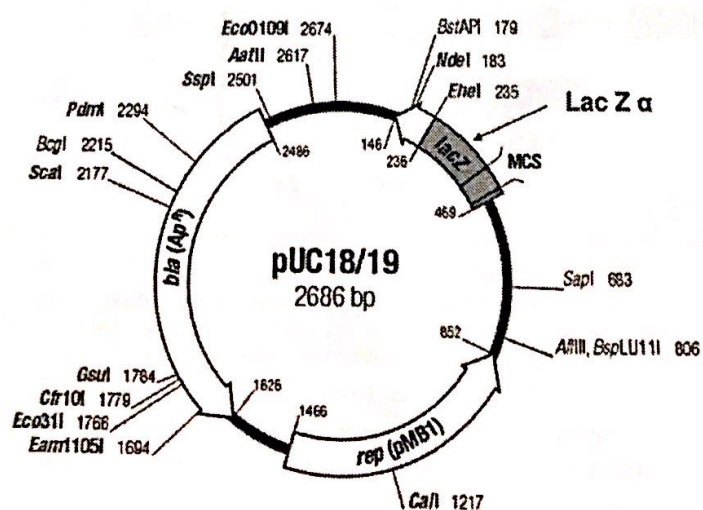
7. **DNA sequencing primers** (we have to sequence a lot of types to know whether we sequenced the target gene or not)

Cloning plasmids



Polylinker site; is a site that contain many restriction sites (up to 20)

The pUC 18 or 19 Cloning Vector



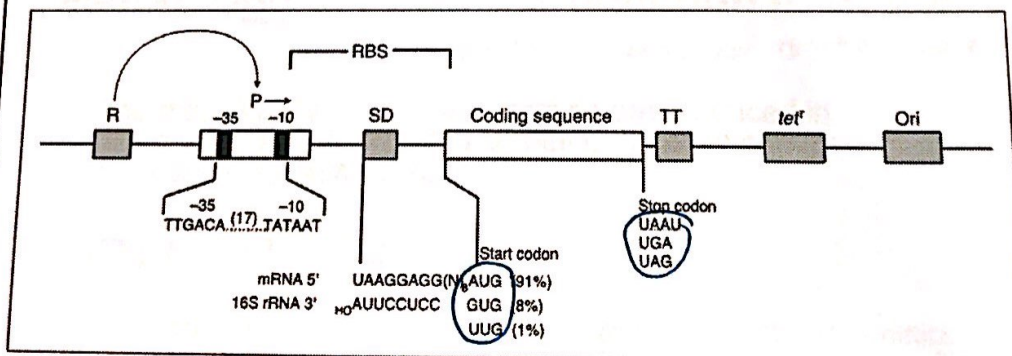
RE Sites in blue occur only once in the plasmid

Expression vectors

- The gene/cDNA coding for a potential target protein must be expressed at high levels.
- This process involves insertion of the gene into a vector that will support high-level transcription and translation (**Expression vector**)
- A wide range of such expression vectors are now commercially available and obviously each is tailored to work best in a specific host cell type (e.g. bacterial, yeast, mammalian).

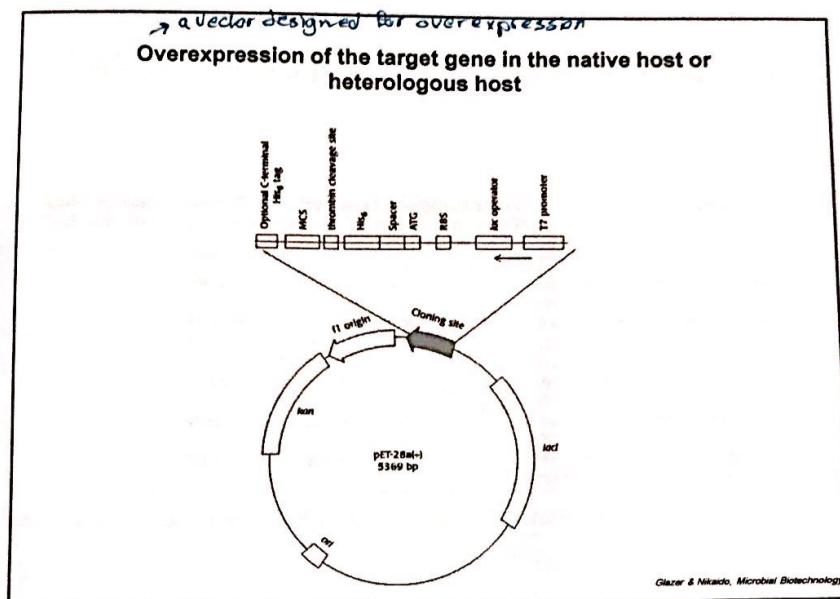
prokaryotes ← → eukaryotes
 * vectors that work in bacteria are different from the ones that work in yeast and in mammalian cells, etc.

Expression vectors



- (P): *tac* promoter consisting of the -35 and -10 sequences, which are separated by a 17-base spacer.
- (RBS): The ribosome binding site consists of the Shine-Dalgarno (SD) sequence followed by an AT-rich translational spacer that has an optimal length of approximately eight bases. The SD sequence interacts with the 3' end of the 16S ribosomal RNA during translational initiation (certain features that has to be there)
- Start codons: with the frequency of their usage in *E. coli*.
- Stop codons: UAA followed by U is the most efficient translational-termination sequence in *E. coli*.
- (R): repressor encoded by a regulatory gene (R) that modulates the activity of the promoter.
- (TT): The transcription terminator serves to stabilize the mRNA and the vector
- *tetr*: an antibiotic-resistance gene (e.g. for tetracycline) facilitates phenotypic selection of the vector
- *Ori*: the origin of replication (*Ori*) determines the vector copy number.

Hannig and Makrides (1998) focus



Heterologous protein production in *E. coli*

→ a protein that is not synthesized inside the bacteria naturally

- The majority of recombinant proteins are produced in microbial systems, with *E. coli* being the most common expression system used.
- **Why *E. coli*?**
 1. Model organism for studying of prokaryotic genetics, the genetics of *E. coli* is known than any other microorganism. (the most studied microorganism is *E. coli*) (*E. coli* divide every 20 min., most of it is non pathogenic)
 2. Suitable plasmids are available, as were a variety of appropriate powerful and inducible promoters.
- The majority of proteins synthesized naturally by *E. coli* are intracellular and the majority of heterologous proteins expressed in *E. coli* accumulate in the cell cytoplasm, where they can represent 25% or more of total cellular protein.

extracellular
~~intracellular~~

secreted outside, easy to purify

intracellular
~~extracellular~~

difficult to purify because it needs to break the cell and will lose a lot of proteins in progress

Heterologous protein production in *E. coli*

Table 3.4 Some promoters that have been used to control expression of recombinant proteins in engineered *E. coli* cells. For example, the widely used pET expression vectors use a T7-based promoter.

Promoter	Method of Induction	Comment
ara	Addition of arabinose to culture medium	Rapid induction and tight regulation, but repressed by glucose
cad	Adjustment of media to an acidic pH value	High expression levels achieved, cheap inducer
lac	Addition of isopropyl- β -D-thiogalactoside (IPTG) to media; high temperature	Low-level expression, leaky expression under non-inducing conditions
tac	Addition of IPTG to media; high temperature	Well-characterized, high-level but leaky expression
trp	Addition of indoleacrylic acid to media	Well-characterized, high-level but leaky expression
T7	Addition of IPTG to media; high temperature	High-level expression but can have leaky expression under non-inducing conditions

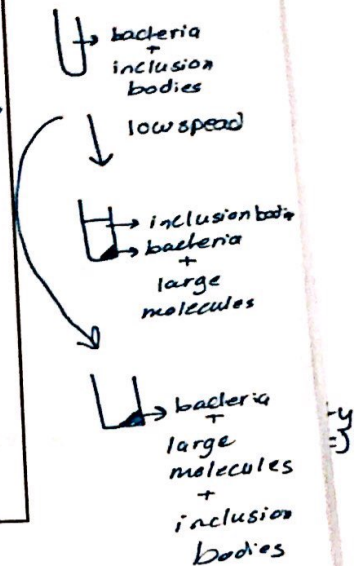
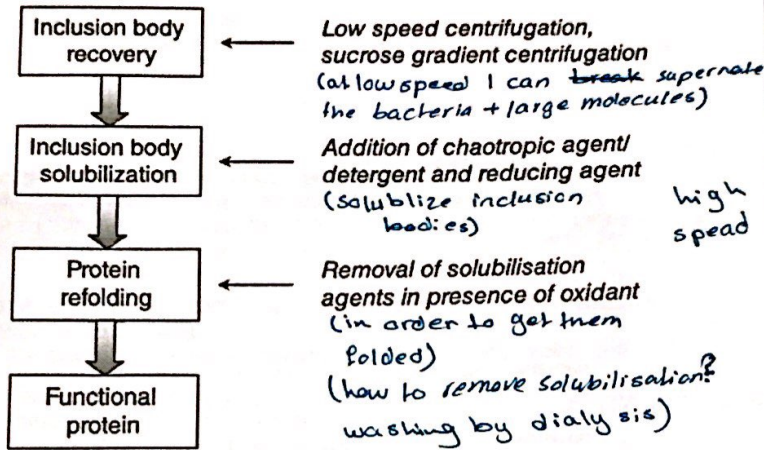
*how do I get the promoter to work? (methods of induction)

Inclusion body formation

→ accumulation of material inside the cell

- Proteins expressed in *E. coli* sometimes accumulates in the cytoplasm in the form of insoluble aggregates termed inclusion bodies
- materials inside the cell
- Inclusion bodies are partially folded intermediates (often exhibiting stretches of intact secondary structure but little or no tertiary structure), and which are therefore obviously functionally inactive. * Inclusion bodies are not fully active proteins (proteins that are produced at a high level, so they aggregate and were not folded correctly that's why they are functionally inactive)
- Inclusion body formed because of
 - Very high local concentrations of the expressed protein in the cytoplasm which may lead to precipitation (high expression is not always good)
 - The presence of insufficient chaperones/folding enzymes (chaperones help the protein)
 - Prevention of formation of disulfide linkages due to the reducing environment of the cytoplasm (covalent bonds)
 - Lack of post-translational modifying enzymes may yield less stable (eukaryotic) (chaperones do post translation modification)

Recovery of proteins from Inclusion body



Production of soluble heterologous protein in *E. coli*

1. Empirical determination of exact optimal expression system: the host strain, plasmid, promoter sequence, . . etc.. *→ a must*
2. Growth of recombinant cells below optimal growth temperatures: this slows the growth and division rate of the cells, thereby also reducing the rate of recombinant protein synthesis.
3. Enhancing the endogenous production of a cofactor (or exogenously providing the cofactor) can increase soluble yields of a recombinant protein which requires the cofactor concerned.
4. Co-expression of chaperones can improve soluble yield of heterologous protein.
5. Expression of the desired product as a fusion protein

attach it to a protein that's highly solubilized

Fusion Partners

* they are not universal, not one size fits all

- Fusion to a (highly soluble) native host cytoplasmic protein enhances otherwise insoluble heterologous proteins. (enhance solubility)
- In addition to that, expression levels of the recombinant protein construct increases.
- Two of the more common fusion partners are maltose-binding protein (a 42-kDa protein encoded by the *malE* gene) and thioredoxin A (a 116-kDa cytoplasmic, highly soluble oxidoreductase enzyme). (so the target protein will be attached to the fusion protein) ^{which are very soluble}
- Fusion partner approach does suffer from a number of potential drawbacks, including:
 1. No fusion partner works universally to solubilize all proteins, and some proteins will remain insoluble no matter which tag is attached
 2. Subsequent proteolytic tag removal may not be efficient (due to perhaps steric considerations); (the attached proteins might be hard to remove)
 3. The protein may not remain soluble once the tag is removed.

Fusion Partners

Table 2. Fusion partners and their ligands*

Fusion partner	Ligand/matrix
FLAG peptide (DYKDDDDK)	AntiFLAG monoclonal antibodies
His ₆	Ni ²⁺ -nitrilotriacetic acid
Glutathione-S-transferase	Glutathione-sepharose
Staphylococcal protein A	IGG-sepharose
Streptococcal protein G	Albumin
Calmodulin	Organic and peptide ligands, DEAE-sepharose
Calmodulin-binding peptides	Calmodulin
Thioredoxin	ThioBond™ resin
β -Galactosidase	TPC®-sepharose
Ubiquitin	Chloramphenicol-sepharose
Chloramphenicol acetyltransferase	Sprotein (ribonuclease A, residues 21-124)
Sprotein (ribonuclease A, residues 1-20)	
Myosin heavy chain	
DsbA	
Biotin subunit (in vivo biotinylation)	
Biotin	Biotin
Streptavidin	Biotin
Streptag	Streptavidin
c-Myc	Anti-Myc antibody
Dihydrofolate reductase	Methotrexate-agarose
PKS	
Polyarginine	S-Sepharose
Polycystine	Thiopropyl-sepharose
Polyphenylamine	Phenylsuperose
lac Repressor	lac Operator
T4 gp35	
Growth hormone, N-terminus	
Maltose-binding protein	Amylose resin
Galactose-binding protein	Galactose-sepharose
Cyclomaltodextrin glucanotransferase	α -Cyclodextrin-agarose
Cellobiose-binding domain	Cellobiose
Haemolysin A (E. coli)	
α CI Protein	
TrpE or TrpLE	
Protein-kinase site(s)	
WAb-Trp-Trp-Pro, WAb epitope	
BTAG	
VP7 protein region of Rotavirus virus	AntiBTAG antibodies
Green fluorescent protein	

*In addition to their utility in purification and detection, specific fusion peptides may confer advantages on the target protein during expression, such as increased solubility or yield, protection from proteolysis, improved folding, and secretion. The engineering of specific primase sites in fusion proteins facilitates the cleavage and removal of the fusion partner(s).

*TPC, p-aminophenyl-p-mercaptoacetate.

*CTP, 3-deoxy-5-carboxymethyluridine cytosine triphosphate.

*Haemolysin, haemagglutinin.

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Hannig and Makrides (1998) focus

Extracellular production

- Advantages of extracellular production of recombinant proteins in *E. coli*:

1. Ease of subsequent protein recovery and simplification of subsequent protein purification as *E. coli* exports very few proteins (they are secreted so we'll find them in the media)
2. Reduced likelihood of proteolytic degradation of the recombinant protein (*E. coli* produces many intracellular proteases but few extracellular proteases);
3. Increased likelihood of correct protein disulfide bond formation (disulfide bonds will not form in the cytoplasm due to its reducing chemical environment).

→ the lysozyme in its normal state inside the undamaged cell will only work when needed but once it's outside the broken cell it'll destroy anything it encounter

* can be performed without damaging the cells, easier

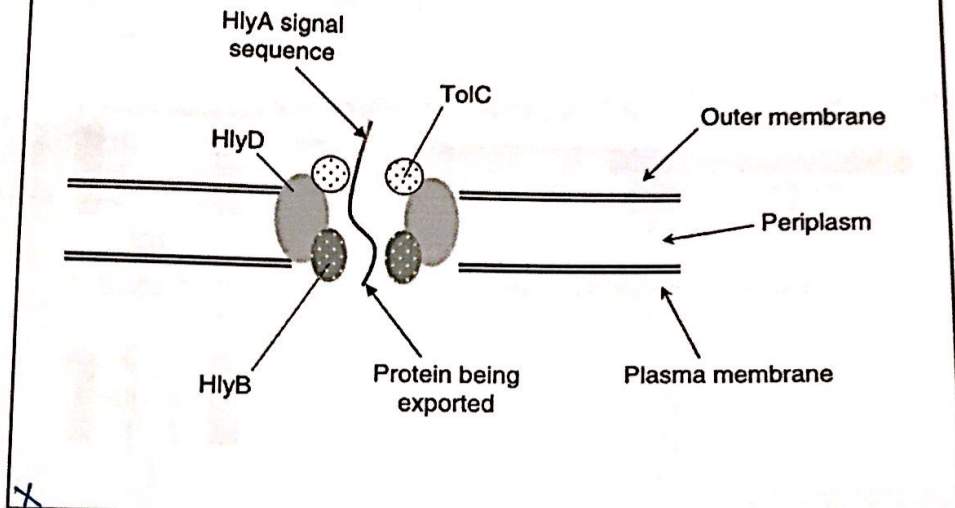
* extracellular means that the bacteria should secrete it in a way

Extracellular production

- Extracellular protein export from Gram-negative bacteria is complex in that it requires translocation through two membranes.
- At least five different secretory transport systems have been identified
- Type I and type II secretory systems predominate in *E. coli*, and both have been used in attempts to develop engineered strains capable of exporting recombinant protein extracellularly.
- The type I secretory mechanism achieves a one-step transport of exported proteins across both membranes
- (how they are exported)
- Native proteins secreted via this system have a characteristic C-terminal targeting signal peptide sequence.
- This signal sequence can be appended to recombinant proteins in order to facilitate their export via this transport mechanism.

→ fusion protein

The Type I secretory system

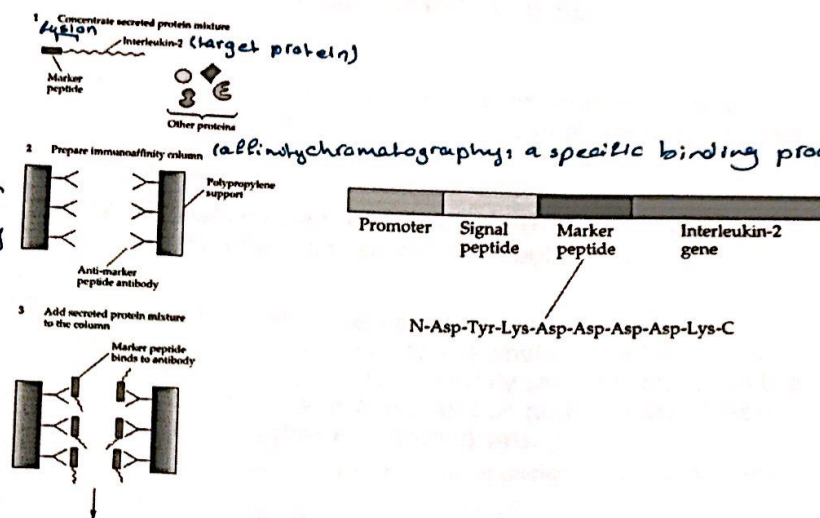


Extracellular production

- Disadvantages of extracellular recombinant protein production from *E. coli*:

1. High-level recombinant protein expression usually overwhelms the secretory systems
(more than the secretion pores can handle)
2. The signal peptide can remain attached to the protein on its export (usually necessitating its subsequent removal from the purified recombinant protein).
→ then it needs an extra enzymatic digestion step to help it detach
3. The yield of true extracellular product is less than 1 g/L of fermentation media. In contrast, intracellular expression will often generate protein levels of 5–10g/L.

Genetic construct used to produce a secreted fusion protein



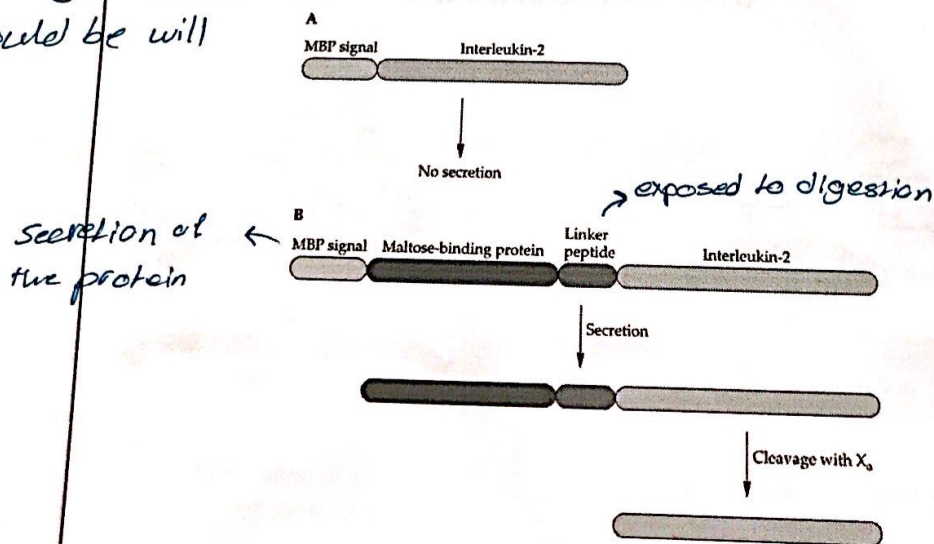
this will recognize the interleukin-2 + marker peptide only and will eliminate other proteins

ex. enzyme-substrate
antibody-antigen

then we add eluent that'll break the bond between Ab-Ag to get the pure protein

then we'll separate the bond between the protein and marker peptide using a specific chemical material (ex. endonucleases) to get the target protein alone

Genetic construct used to produce a secreted fusion protein



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Integration of a cloned gene into a chromosomal site

In some cases, it is necessary to produce proteins from a cloned genes integrated into a chromosomal site to overcome the following problems:

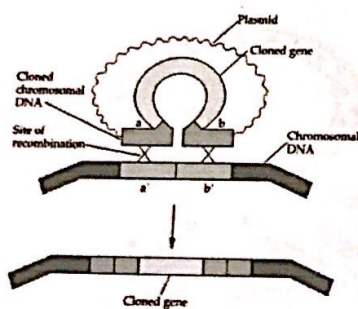
1. Avoid plasmid loss during cell growth, especially during pilot plant- or industrial-scale fermentations
2. For genetically engineered microorganisms that are designed to be released into the environment to remain both effective and environmentally safe, it is essential that the cloned DNA be retained and be neither easily lost nor transferred to other microorganisms.

(bacteria can't use its genome to another bacteria while it's alive)

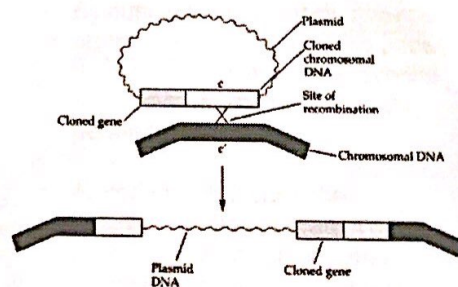
* the cell without plasmid (the process of losing plasmid) is called curing plasmid, how? applying an electric current that'll open the pores and allow the plasmid to get out

by c)

Integration of a cloned gene into a chromosomal site



Integration of the cloned gene only



Integration of the whole plasmid

(depends on how many other genes there's in the plasmid)

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Table 3.6 Some advantages and disadvantages of *E. coli* used for the production of recombinant proteins.

Advantages

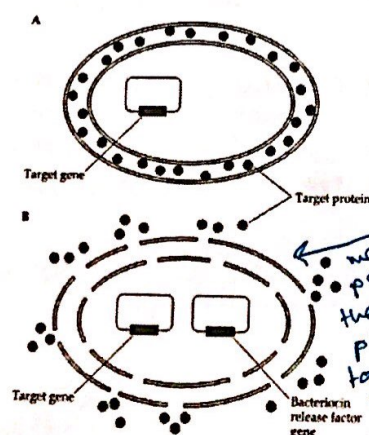
E. coli genetics well characterized, allowing straightforward genetic manipulation
 Expression vectors available which facilitate straightforward and high-level recombinant protein expression
 Grows rapidly and to high cell densities on relatively inexpensive fermentation media
 Suitable fermentation technology well established
 History of safe use for the production of many commercialized therapeutic proteins (ex. ~~insulin~~ → insulin)

Disadvantages

Poor ability to secrete recombinant proteins extracellularly in large amounts
 Recombinant protein often accumulates intracellularly in inactive form (as inclusion bodies)
 Unable to undertake post-translational modification of proteins
 Presence of an additional methionine residue at the N-terminal of the recombinant protein (not easily post-translational modification)

* golgi apparatus adds groups and do post-translational modification

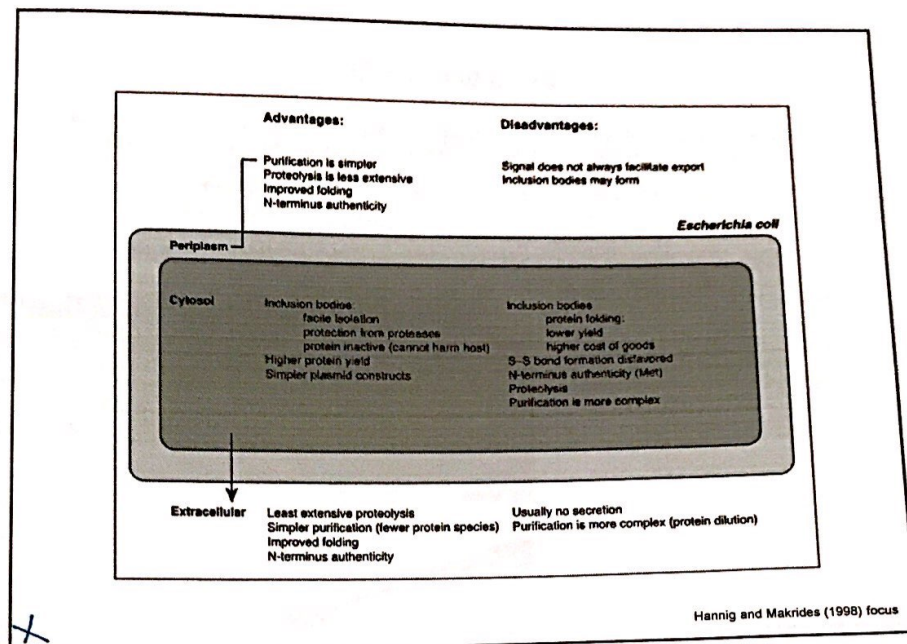
***E. coli* cells can be engineered to secrete a foreign protein to the periplasm**



• By putting the bacteriocin release protein gene onto a plasmid under the control of a strong regulatable pro-moter, *E. coli* cells may be permeabilized at will.

• *E. coli* cells that carry the bacteriocin release protein gene are transformed with another plasmid carrying a cloned gene that has been fused to a secretion signal peptide sequence.

* proteins are too small compared to a plasmid



Heterologous production in bacteria other than *E. coli*

- Other bacteria can be used as expression systems. Bacteria which are listed as GRAS (generally recognized as safe) serve as alternative production systems.
- Heterologous proteins are produced in Gram-positive species of *Streptomyces* and bacilli. These are non-pathogenic, well characterized bacteria and secrete large quantities of protein extracellularly.
- Approximately, 50% of commercially available industrial enzymes are produced by bacilli spp.
- Considerable experience has also accumulated with regard to their industrial-scale culture and their genetic manipulation.

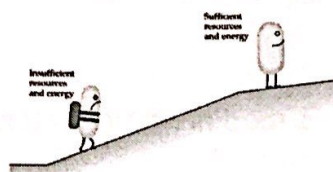
*type strain, most studied strain, very well characterized, has to be GRAS

*streptomyces are soil bacteria, most of them are GRAS

overloading the bacteria
due to the introduction of
a new DNA

Metabolic Load

- Expression of foreign DNA in a host organism often change the metabolism of the organism in ways that may impair normal cellular functioning. This phenomenon, is due to a metabolic load (metabolic burden; metabolic drain) that is imposed upon the host by the foreign DNA and can occur because of the following.
 1. An increasing plasmid copy number and/or size requires increasing amounts of cellular energy for plasmid replication and maintenance.
 2. The limited amount of dissolved oxygen in the growth medium
 3. Overproduction of both target and marker proteins may deplete the pools of certain aminoacyl-tRNAs (or even certain amino acids) and/or drain the host cell of its energy (in the form of ATP or GTP).
 4. When a foreign protein is overexpressed and then exported from the cytoplasm to the cell membrane, the periplasm, or the external medium, it may "jam" export sites and thereby prevent the proper localization of other, essential host cell proteins.
 5. The foreign protein per se may interfere with the functioning of the host cell.



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it needs certain amino acids for its proteins and now it needs more for the new proteins

Heterologous protein production in yeast

Table 3.8 Some advantages and disadvantages of heterologous protein production in yeast.

Advantages

Most are GRAS listed (ex. *saccharomyces cerevisiae*)
Proven history of use in many biotechnological processes
Fermentation technology is well established
Ability to carry out post-translational modifications of recombinant proteins (because it's a prokaryote)

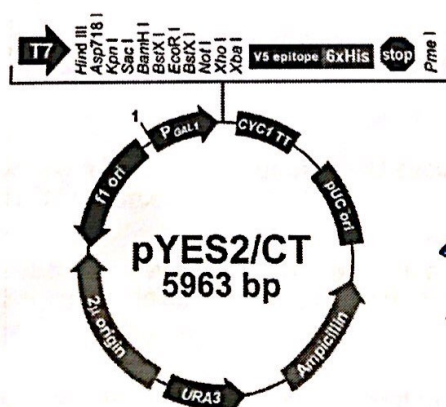
Disadvantages

Recombinant proteins usually expressed at relatively low levels
Retention of many exported proteins in the periplasmic space
Some post-translational modifications differ significantly from those achieved by animal cells

Saccharomyces cerevisiae

- There are three main classes of *S. cerevisiae* expression vectors:
 → self replication (independent)
 - Yeast episomal plasmids (YEPs).
 - Yeast integrating plasmids (YIps)
 - Yeast artificial chromosomes (YACs)
- Yeast episomal plasmids have been used extensively for the production of either intra- or extracellular heterologous proteins.
- Typically, vectors function in both *E. coli* and *S. cerevisiae*.

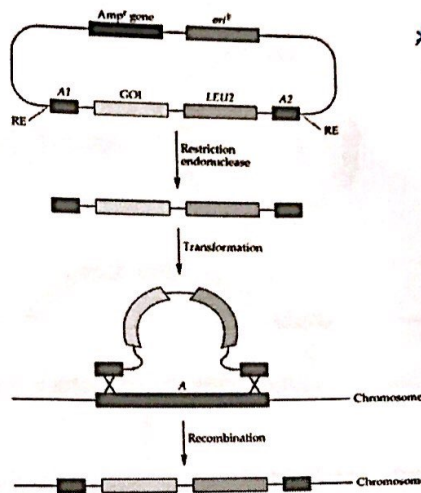
Heterologous protein production in yeast by YIps



← vectors used in yeast
these vectors replicate independently

URA3:	Auxotrophic marker
CYC1 TT:	Terminator
Amp ^R :	Ampicillin resistance gene
pUC ori:	origin of replication (<i>E. coli</i>)
2μ:	origin of replication (Yeast)
CEN:	Centromere is required for segregation of chromosomes during mitosis
ARS:	is the origin of replication in the yeast genome

Integration of DNA with a Yip vector



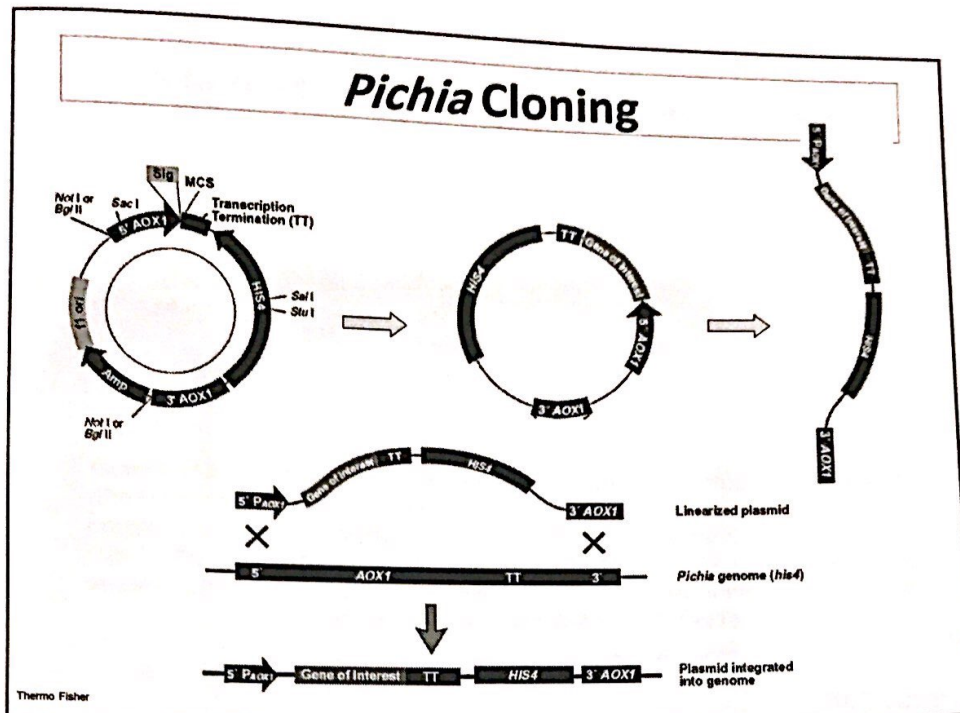
* this method can be used to solve antibiotic resistance

* the integration will interrupt the gene

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Heterologous production in yeast → producing for commercial purposes

- While *Saccharomyces cerevisiae* remains the most popular yeast host, several other yeasts are also utilized in the production of heterologous proteins.
- Pichia pastoris* is among a limited number of yeasts capable of utilizing methanol as sole carbon source.
- Its methanol utilization pathway contains two alcohol oxidase genes (*AOX1* and *AOX2*) and *AOX1* protein levels in particular can reach over 30% of total soluble protein in the cell. these are powerful genes
- AOX1* expression is under the control of a strong and tightly regulated *AOX1* promoter, and this promoter is used in recombinant systems to drive high-level protein synthesis
- Such recombinant systems are also usually designed so as to incorporate a protein export sequence on the recombinant protein, thus ensuring its extracellular export which simplifies subsequent protein recovery.

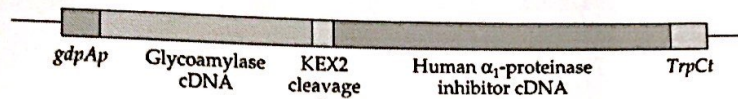


Heterologous protein production in fungi

- Fungi such as *Aspergillus niger* and *Aspergillus oryzae* naturally produce various extracellular enzymes at very high levels (up to 20g/L) and many of these enzymes (cellulases, amylases, proteases, phytases and lipases) have been commercialized. (used in juice industries)
- Advantages of using fungi for heterologous gene expression:
 - Many are GRAS listed and many produce non-recombinant enzymes as well as other primary and secondary products of metabolism (e.g. vitamins, various organic acids, antibiotics and alkaloids). where do we find them? in detergents
how are they produced? by hyperthermal microorganisms (isolated from hot springs)
 - Large-scale fermentation systems are developed and optimized.
 - They have an unparalleled natural capacity to secrete large quantities of certain proteins into the extracellular medium.
 - They possess the enzymatic capability to carry out PTMs (although, their glycosylation capacity differs from that characteristic of mammalian cells)

* extracellular are easier to purify

Heterologous protein production in fungi



Construct for expression and secretion of the human α_1 -proteinase inhibitor in the filamentous fungus *A. niger*. The expression cassette includes the strong constitutive promoter *gdpAp*, the transcriptional terminator from the *TrpC* gene (*TrpCt*), the cDNA encoding glucoamylase to facilitate secretion, and the coding sequence for the Kex2 recognition site for in vivo removal of the glucoamylase fusion protein by the host Kex2 endoprotease.

← produced continuously

Glucosylase cDNA could be a fusion protein to facilitate secretion

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